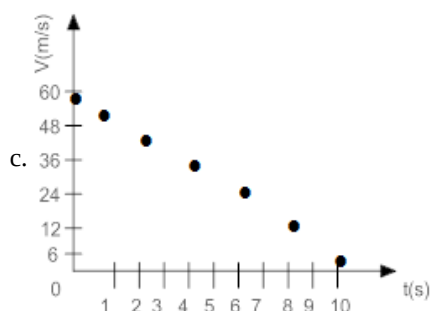


Solution

MOTION-02

Class 09 - Science

1. Velocity of an object is uniform if it travels equal displacement in equal intervals of time.
2. If the velocity of an object decreases, then the object is said to be moving with negative acceleration.
3. Velocity is defined as the ratio of the displacement to the time taken.
4. Displacement is the quantity which is measured by the area under velocity time graph.
5. Both (a) and (b) are false with respect to the concept of displacement.
 - a- displacement can be zero.
 - b- Its magnitude can't be greater than the distance traveled by the object.
6. The graph is a parabola.
7. No, because the speed of a body is the ratio of distance and time and distance travelled is never negative.
8. The object in motion is considered to be a point object if the distance it travels is very large as compared to the dimensions of the object.
9. Average velocity is the ratio of the total displacement to the total time taken.
10. Total distance travelled by the object = 16 m + 16 m = 32 m
Total time taken = 4 s + 2 s = 6 s
Average speed = $\frac{\text{total distance travelled}}{\text{total time taken}}$
$$v_{av} = \frac{s_1 + s_2}{t_1 + t_2} = \frac{(16+16)m}{(4+2)s} = \frac{32}{6} = 5.33\text{m/s}$$
11. acceleration = 2.86 m/s²
time = 30.8s
12. Linear velocity = Angular velocity × Radius of circular path.
13. We need to specify a reference point called the origin.
14. Straight line, passing through the origin is the nature of the displacement time graph of a body moving with constant velocity
15. When the object has uniform motion then displacement and distance have the same magnitude.
16. The reference point from which the distance of a body is measured is called origin.
17. The area occupied below the velocity-time graph measures the distance covered by any object.
18. Acceleration is defined as the change in velocity per unit time.
19. Graphically the path of an object will be linear i.e. look like a straight line when it is in uniform motion.
20. The odometer measures the distance travelled of the automobile.
21. A car travelling along a straight road having much traffic increases or decreases its speed by unequal amounts in equal intervals of time.
22. If the velocity of an object decreases, then the object is said to be moving with negative acceleration. It is also called retardation.
23. When the change in velocity of a body takes place in the direction of motion of the body, then the acceleration is positive.
24. S.I. unit of acceleration is metre / second² (ms⁻²).
25. A fixed point with respect to which the given object changes its position.
26. a. (a) Acceleration = slope of the velocity time graph
$$a = \frac{V_2 - V_1}{t_2 - t_1}$$
$$a = \frac{54 - 24}{1 - 6} = \frac{30}{-5} = -6\text{m/s}^2$$
 - b. Distance = $S = ut + \frac{1}{2}at^2$
$$= 60 \times 10 + \frac{1}{2}(-6) \times (10)^2$$
$$= 600 - 300 = 300\text{m}$$



27. Given, initial velocity of electron, $u = 5 \times 10^4 \text{ ms}^{-1}$ and acceleration, $a = 10^4 \text{ ms}^{-2}$

i. Final velocity of electron $= v = 2 \times u = 2 \times 5 \times 10^4 \text{ ms}^{-1}$

We know that, $v = u + at$ or $t = \frac{v-u}{a} = \left(\frac{10 \times 10^4 - 5 \times 10^4}{10^4} \right) = \frac{5 \times 10^4}{10^4} = 5 \text{ s}$

ii. Using relation; $S = ut + \frac{1}{2}at^2$

Distance covered by electron in the given time, $S = (5 \times 10^4) \times 5 + \frac{1}{2}(10^4) \times (5)^2 = (25 \times 10^4 + \frac{25}{2} \times 10^4) = 37.5 \times 10^4 \text{ m}$

28. i. Velocity-time graph is a straight line. Therefore, it represents uniformly accelerated motion.

ii. Velocity-time graph is a straight line parallel to the time axis. It represents uniform motion.

iii. Velocity-time graph is a straight line having a negative slope. It represents uniformly retarded motion.

iv. Velocity-time graph is a curve having a negative slope. It represents non-uniformly retarded motion.

29. In the first case:

Since body starts from stationary position,

$\therefore$ Initial velocity, $u = 0$

final velocity, $v = 6 \text{ ms}^{-1}$

time, $t = 30 \text{ s}$

$$a = \frac{v-u}{t}$$

Substituting the given values of u , v , and t in the above equation. we get

$$a = \frac{(6 \text{ ms}^{-1} - 0 \text{ ms}^{-1})}{30 \text{ s}}$$

$$= 0.2 \text{ ms}^{-2}$$

In the second case:

initial velocity, $u = 6 \text{ ms}^{-1}$

time, $t = 5 \text{ s}$

$$\text{Then, } a = \frac{(4 \text{ ms}^{-2} - 6 \text{ ms}^{-2})}{5 \text{ s}}$$

$$= -0.4 \text{ ms}^{-2}$$

The acceleration of the bicycle in the first case is 0.2 ms^{-2} and in the second case, it is -0.4 ms^{-2}

30. On taking the formula when both bodies are moving along same direction, we have

(a) Relative velocity of A with respect to B is -

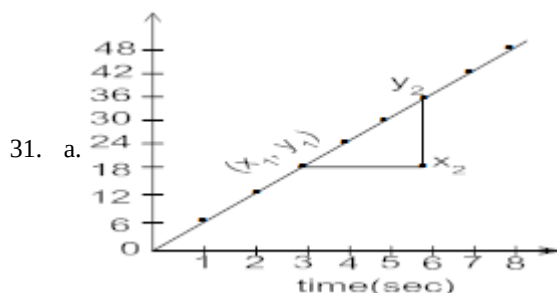
$$V_{AB} = V_A - V_B = 80 - 50 = 30 \text{ km/h}$$

Hence, the velocity of car A with respect to car B is 30 km/h in same direction.

(b) Relative velocity of B with respect to A is -

$$V_{BA} = V_B - V_A = 50 - 80 = -30 \text{ km/h}$$

Hence, the velocity of car B with respect to car A is 30 km/h in opposite direction.



b. Slope of the displacement - time graph = velocity

$$\text{c. velocity} = \frac{48-0}{8-0} = 6 \text{ m/s}$$

32. Height = h

Distance = $s = 320 \text{ m}$

Acceleration due to gravity = $g = 10 \text{ m/s}^2$

Initial velocity = $u = 0$

(i) from $s = ut + \frac{1}{2}at^2$

$$h = ut + \frac{1}{2}gt^2$$

$$320 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\frac{320 \times 2}{10} = t^2$$

$$64 = t^2 \quad t = \sqrt{64}$$

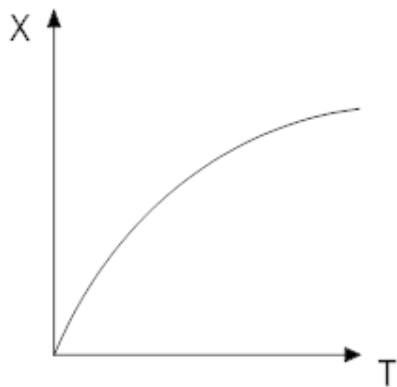
$$t = 8 \text{ sec}$$

$$(ii) \text{ from } v = u + at$$

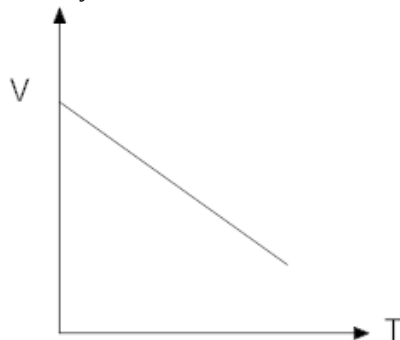
$$v = 0 + 10 \times 8$$

$$v = 80 \text{ m/s}$$

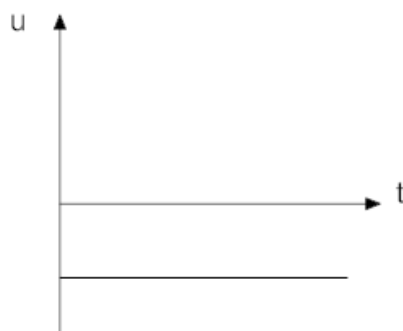
33. i. Position – time



ii. Velocity – time



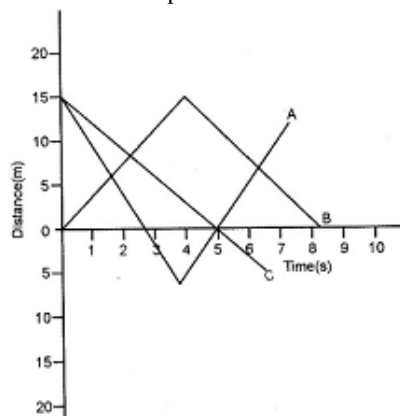
iii. Acceleration- time



34. i. GRAPH A: The displacement of the body is 12 m and the distance travelled is 37 m

ii. GRAPH B: Displacement is zero and the distance travelled is $15 + 15 = 30$ m

iii. GRAPH C: Displacement is about 5 m and distance travelled is about 20 m.



Graph C represents a motion with negative acceleration.

35. a. We know that area under v-t graph gives displacement.

So, Area = distance (S) = area of triangle + area of rectangle

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 6 \times 10$$

$$= 30 \text{ m}$$

$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

$$= (16-6) \times 10$$

$$= 10 \times 10$$

$$= 100 \text{ m}$$

$$\text{Total area} = 100 + 30 = 130 \text{ m}$$

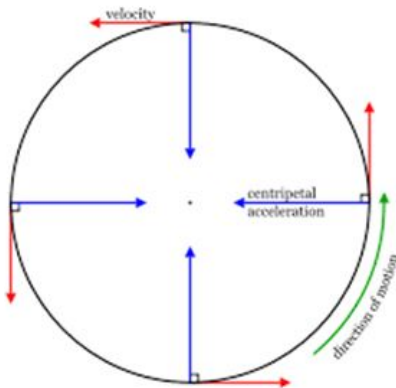
$$\text{Therefore distance covered by the runner in 16s} = 130 \text{ m}$$

b. Since, at $t = 11$ sec, runner is travelling with uniform velocity so, there is no change in velocity hence acceleration is zero.

36. (a) An object with a constant acceleration can still have the zero velocity. For example an object which is at rest on the surface of earth will have zero velocity but still being acted upon by the gravitational force of earth with an acceleration of 9.81 ms^{-2} towards the center of earth. Hence when an object starts falling freely can have constant acceleration but with zero velocity.

(b) When an athlete moves with a velocity of constant magnitude along the circular path, the only change in his velocity is due to the change in the direction of motion. Here, the motion of the athlete moving along a circular path is, therefore, an example of an accelerated motion where acceleration is always perpendicular to direction of motion of an object at a given instance. Hence, it is possible when an object moves in a circular path.

37. When a body moves in a circle, it is called circular motion. When the velocity of an object changes, we say that the object is accelerating. The change in the velocity could be due to change in its magnitude or the direction of the motion or both.



If the athlete is running along a hexagonal shaped path ABCDEF, the athlete will have to change his direction six times while he completes one round. If the athlete moves with a velocity of constant magnitude along the circular path, the only change in his velocity is due to the change in the direction of motion.

The motion of the athlete moving along a circular path is an example of an accelerated motion.

The circumference of a circle of radius r is given by $2\pi r$. If the athlete takes t seconds to go once around the circular path of radius r , the velocity v is given by

$$v = \frac{2\pi r}{t}$$

When an object moves in a circular path with uniform speed, its motion is called uniform circular motion.

38. Speed of the cheetah in ms^{-1} is $90 \times \frac{5}{18} \text{ ms}^{-1} = 25 \text{ ms}^{-1}$

Distance of the prey = 100 m

$$\text{Therefore time to catch the prey} = \frac{\text{Distance}}{\text{Speed}} = \frac{100}{25} = 4 \text{ s}$$

39. Initial speed = $u = 72 \text{ km/hr}$

$$= \frac{72 \times 5}{18} = 20 \text{ m/s}$$

Final speed = $v = 54 \text{ km/h}$

$$= \frac{54 \times 5}{18} = 15 \text{ m/s}$$

Distance = $s = 70 \text{ m}$

$$\text{Now, } v^2 - u^2 = 2as$$

$$(15)^2 - (20)^2 = 2 \times a \times 70$$

$$225 - 400 = 140a$$

$$-175 = 140a$$

$$a = -1.25 \text{ m/s}^2 \text{ (negative sign shows retardation)}$$

40. The shape of the path followed by the ball is parabolic.

Vertically downward motion is due to gravitational acceleration. Horizontal motion is due to zero acceleration.

41. $u = 5 \text{ ms}^{-1}$, $a = -10 \text{ ms}^{-2}$

$v = 0$ (since at maximum height its velocity will be zero)

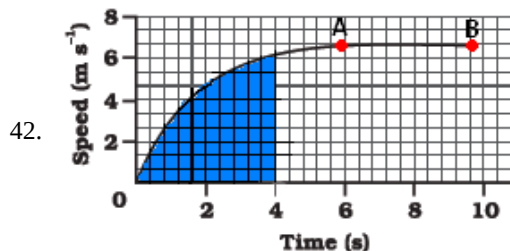
$$v = u + at = 5 + (-10) \times t$$

$$0 = 5 - 10t$$

$$10t = 5, \text{ or, } t = 5/10 = 0.5 \text{ second.}$$

$$s = ut + \frac{1}{2}at^2 = 5 \times 0.5 + \frac{1}{2} \times (-10) \times 0.5^2$$

$$= 2.5 - 1.25 = 1.25 \text{ m}$$



a. Distance travelled by car in the 4 second

The area under the slope of the speed – time graph gives the distance travelled by an object.

In the given graph

56 full squares and 12 half squares come under the area slope for the time of 4 seconds.

$$\text{Total number of squares} = 56 + \frac{12}{2} = 62 \text{ squares}$$

The total area of the squares will give the distance travelled by car in 4 seconds. on the time axis,

5 squares = 2 seconds, therefore 1 square = $\frac{2}{5}$ seconds

on speed axis there are 3 squares = 2 m/s

$$\text{therefore, area of one square} = \frac{2}{5} \text{ s} \times \frac{2}{3} \text{ s} = \frac{4}{15} \text{ m}$$

$$\text{so area of 62 squares} = \frac{4}{15} \text{ m} \times 62 = \frac{248}{15} \text{ m} = 16.53 \text{ m}$$

Hence the car travels 16.53 m in the first 4 seconds.

b. The straight line part of graph, from point A to point B represents a uniform motion of car.

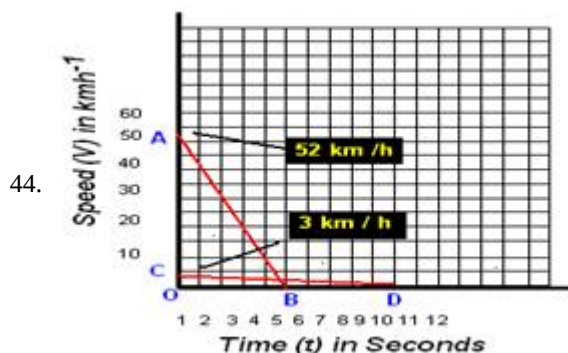
43. Given speed of the bus = $60 \text{ kmh}^{-1} = \frac{5}{18} \times 60 = 16.7 \text{ ms}^{-1}$

$$\text{Time of reaction} = \frac{1}{15} \text{ s}$$

$$\text{Time of reaction under the influence of alcohol} = \frac{1}{2} \text{ s}$$

i) Distance travelled by the bus in the first case distance = Speed $\times$ time = $16.7 \times \frac{1}{15} = 1.11 \text{ m}$

ii) Distance travelled by the bus in the second case distance = Speed $\times$ time = $16.7 \times \frac{1}{2} = 8.35 \text{ m}$



As given in the figure below AB (in red line) and CD (in red line) are the Speed-time graph for given two cars with initial speeds 52 kmh^{-1} and 3 km h^{-1} respectively.

Distance Travelled by first car before coming to rest = Area of $\triangle OAB$

$$= \left(\frac{1}{2}\right) \times OB \times OA$$

$$= \left(\frac{1}{2}\right) \times 5 \text{ s} \times 52 \text{ kmh}^{-1}$$

$$= \left(\frac{1}{2}\right) \times 5 \times (52 \times 1000)/3600 \text{ m}$$

$$= \left(\frac{1}{2}\right) \times 5 \times \left(\frac{130}{9}\right) \text{ m}$$

$$= \frac{325}{9}$$

$$= 36.11 \text{ m}$$

Distance Travelled by second car before coming to rest = Area of $\triangle OCD$

$$= \left(\frac{1}{2}\right) \times OD \times OA$$

$$= \left(\frac{1}{2}\right) \times 10 \text{ s} \times 3 \text{ kmh}^{-1}$$

$$= \left(\frac{1}{2}\right) \times 10 \times \left(\frac{3 \times 1000}{3600}\right) \text{ m}$$

$$= \left(\frac{1}{2}\right) \times 10 \times \left(\frac{5}{6}\right) \text{ m}$$

$$= 5 \times \left(\frac{5}{6}\right) \text{ m}$$

$$= \frac{25}{6} \text{ m}$$

$$= 4.16 \text{ m}$$

$\therefore$ Clearly the first car will travel farther (36.11 m) than the first car (4.16 m).

45. i. Volume in iron ball = 200 cm^3

$$\therefore \text{Mass of iron ball} = 200 \times 10^{-6} \times 7800 = 1.56 \text{ kg}$$

$$\text{Weight of iron ball in air} = m \times g = 1.56 \times 10$$

$$\text{Weight of iron ball in air} = 15.6 \text{ N}$$

ii. Upthrust = Volume of water displaced (V) $\times$ density of water (d) $\times$ g

$$= Vdg = 2 \times 10^{-4} \times 1000 \times 10$$

$$\text{Upthrust} = 2 \text{ N}$$

iii. Apparent weight = True weight - Upthrust = (15.6 - 2)

$$\text{Apparent weight} = 13.6 \text{ N}$$

iv. Apparent density = density of solid - density of liquid = $7800 - 1000$

$$\text{Apparent density} = 6800 \text{ kgm}^{-3}$$

46. a. Distance covered = $\pi \times OA = \pi \times 5 = 5\pi \text{ cm}$

b. Displacement = $2 \times OB$

$$= 2 \times 5 = 10 \text{ cm along AB}$$

47. Initial speed of each train $u = 72 \text{ kmh}^{-1} = 20 \text{ ms}^{-1}$

$$\text{Distance travelled by train A in } 50 \text{ s} = 20 \times 50 = 1000 \text{ m}$$

Distance travelled by train B in 50 s with an acceleration of 1 ms^{-2} is

$$20 \times 50 + \frac{1}{2} (1) \times (50)^2 = 2250 \text{ m using } S = ut + \frac{1}{2} at^2$$

Therefore the original distance between the trains is $2250 - 1000 = 1250 \text{ m}$

48. i. The distance-time graph represents the line AB which shows the speed of the body. So,

$$\text{speed} = \frac{\text{Dis tance}}{\text{Time}}$$

$$= \frac{3 \text{ cm}}{(5-2) \text{ s}}$$

$$= 1 \text{ cm/s}$$

ii. The distance-time graph shows that the body is at rest between graph line B to C, it means no movement. So speed is zero i.e.,

$$\text{speed} = \frac{\text{Dis tance}}{\text{Time}} = \frac{0 \text{ cm}}{(7-5) \text{ s}}$$

iii. The distance-time graph represents the line CD which shows the speed of the body. So,

$$\text{speed} = \frac{\text{Dis tance}}{\text{Time}}$$

$$= \frac{(7-3) \text{ cm}}{(9-7) \text{ s}} = \frac{4 \text{ cm}}{2 \text{ s}}$$

$$= 2 \text{ cm/s}$$

49. Given total distance travelled = $15 + 75 + 10 = 100 \text{ km}$

$$\text{Time taken in the first part of motion } t_1 = \frac{S}{V_1} = \frac{15}{30}$$

$$\text{Time taken in the second part of motion } t_2 = \frac{S}{V_2} = \frac{75}{50}$$

$$\text{Time taken in the third part of motion } t_3 = \frac{S}{V_3} = \frac{10}{20}$$

$$\text{Total time taken } t = t_1 + t_2 + t_3 = \frac{15}{30} + \frac{75}{50} + \frac{10}{20} = 2.5 \text{ h}$$

$$\text{Therefore, } V_{av} = \frac{\text{Total dis tance travelled}}{\text{Total time taken}} = \frac{100}{2.5} = 40 \text{ kmh}^{-1}$$

50. i. Consider the linear motion of a body with initial velocity 'u'. Let the body accelerate uniformly and acquire a final velocity 'v' after time 't'.

$$\text{Then, Average velocity of body} = \frac{\text{Initial velocity} + \text{Final velocity}}{2} = \frac{u+v}{2}$$

$$\therefore \text{The distance covered by the body in time 't' } S = \text{Average velocity} \times \text{Time taken} \Rightarrow S = \frac{u+v}{2} \times t$$

We know that, $v = u + at$

$$\Rightarrow S = \frac{u+(u+at)}{2} \times t$$

$$\text{or, } S = \frac{2ut+at^2}{2}$$

$$\Rightarrow S = ut + \frac{1}{2}at^2 \text{ Which is required equation.}$$

$$\text{ii. We know that, } S = ut + \frac{1}{2}at^2 \dots\dots\dots (1)$$

$$\text{Also, } a = \frac{v-u}{t}$$

$$\Rightarrow t = \frac{v-u}{a}$$

Putting the value of t in (1), we have

$$S = u \left(\frac{v-u}{a} \right) + \frac{1}{2}a \left(\frac{v-u}{a} \right)^2$$

$$\text{or } S = \frac{uv-u^2}{a} + \frac{v^2+u^2-2uv}{2a}$$

$$\text{or } 2aS = 2uv - 2u^2 + v^2 + u^2 - 2uv$$

$$\text{or } v^2 - u^2 = 2aS. \text{ Which is required equation.}$$

51. Here, initial velocity, $u = 90 \text{ kmh}^{-1}$

$$= \frac{90 \text{ km}}{1 \text{ h}}$$

$$= \frac{90 \times 1000 \text{ m}}{60 \times 60 \text{ s}}$$

$$= 25 \text{ ms}^{-1}$$

$$\text{Acceleration, } a = -0.5 \text{ ms}^{-2}$$

$$\text{Final velocity } v = 0$$

$$\text{From the equation of motion, } 2as = v^2 - u^2$$

$$s = \frac{v^2 - u^2}{2a}$$

$$= \frac{0 - (25 \text{ ms}^{-1})^2}{2 \times (-0.5 \text{ ms}^{-2})}$$

$$= \frac{-625 \text{ m}^2 \text{ s}^{-2}}{-1.0 \text{ ms}^{-2}}$$

$$= 625 \text{ m}$$

The train will go 625 m further, after applying the brakes.

52. Speed = Slope of distance - time graph. The smaller the slope, the smaller is the speed.

53. a. $u = 0$, $a = 0.1 \text{ ms}^{-2}$, $t = 2 \text{ min} = 120 \text{ seconds}$.

$$v = u + at = 0 + 0.1 \times 120 = 12 \text{ ms}^{-1}$$

$$\text{so speed acquired} = v = 12 \text{ ms}^{-1}$$

$$\text{b. } S = 0 \times 120 + \frac{1}{2} \times 0.1 \times 120^2 = 720 \text{ m.}$$

54. $u = \text{initial velocity} = 0$ (body starts from rest)

$$S = \text{distance} = 16 \text{ m}$$

$$T = \text{time} = 4 \text{ s}$$

$$\text{(i) From, } s = ut + \frac{1}{2}at^2$$

$$16 = 0 \times t + \frac{1}{2} \times a \times (4)^2$$

$$16 = \frac{1}{2} \times a \times 16$$

$$\frac{16 \times 2}{16}$$

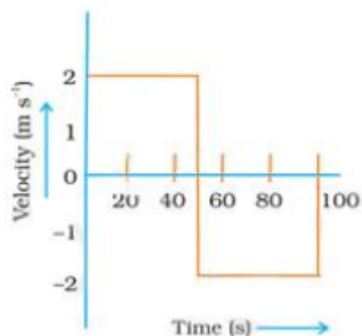
$$a = 2 \text{ m/s}^2$$

$$\text{(ii) From, } v = u + at$$

$$v = 0 + 2 \times 4$$

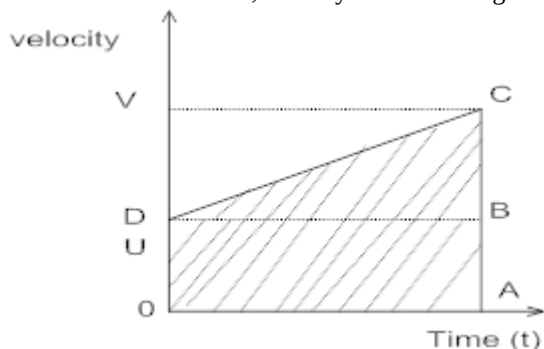
$$v = 8 \text{ m/s}$$

55. For the initial 50 seconds, velocity is 2 m/s. After that, velocity drops of zero; as shown by a vertical line in the graph.



For the next 50 seconds, velocity is taken in negative because displacement is becoming zero.

56.



Let at time $T=0$, body moves with initial velocity u and at time ' t ' body has final velocity ' v ' and in time ' t ' it covers a distance ' S '.

$AC = v$, $AB = u$, $OA = t$, $DB = OA = t$, $BC = AC - AB = v - u$

Area under a v - t curve gives displacement so,

$S = \text{Area of triangle } DBC + \text{Area of rectangle } OABD \dots\dots(i)$

Area of $\triangle DBC = \frac{1}{2} \times \text{Base} \times \text{Height}$

$= \frac{1}{2} \times DB \times BC$

$= \frac{1}{2} \times t \times (v - u) \dots\dots(ii)$

Area of rectangle $OABD = \text{length} \times \text{breadth}$

$= OA \times BA$

$= t \times u \dots\dots(iii)$

From (i), (ii) and (iii)

$S = ut + \frac{1}{2} \times t \times (v - u)$

$S = ut + \frac{1}{2} \times t \times at$

$S = ut + \frac{1}{2}at^2$

57. i. This is impossible

ii. This is possible because when an object is thrown upward, at the highest point its velocity becomes zero for a slice of time, but its acceleration (g) remains due to the attraction of gravity.

iii. It is not possible.

iv. It is possible because when a moving car applies the brake its velocity decreases. The acceleration is against the direction of the motion of the car.

58. First km/hr is converted into m/s for its formula to be applicable in a standard form.

$$v = 90 \times \frac{5}{18}$$

$$v = 25 \text{ m/s}$$

We know that,

$$v = \frac{\text{Displacement}}{\text{Time}}$$

i. $v = 25 \text{ m/s}$ and time is given as 1 min, $t = 60 \text{ sec}$

Displacement = Velocity $\times$ Time

$$= 25 \times 60$$

$$\text{Displacement} = 1500 \text{ m} = 1.5 \text{ km}$$

Thus the distance traveled by Soham in his car in one min is 1500 m or 1.5 km.

ii. b) $v = 25 \text{ m/s}$ and the time is given 1 sec, $t = 1 \text{ sec}$

$$\text{Displacement} = \text{Velocity} \times \text{Time}$$

$$= 25 \times 1$$

$$\text{Displacement} = 25 \text{ m} = 0.025 \text{ km}$$

Thus the distance traveled by Soham in his car in one sec is 25 m or 0.025 km.

59. i. According to given graph, B is 100 m ahead of A.

ii. Speed of B = Slope of PQ = $\frac{150-100}{2-0} = 25 \text{ m/s}$

iii. A and B meet at Q, i.e. 150 m from the origin and 2 s after the starting.

iv. Speed of A = Slope of OQ = $\frac{150}{2} = 75 \text{ m/s}$

$$\text{Difference between speeds} = 75 - 25 = 50 \text{ m/s}$$

v. The speed of both the trains is uniform as s-t graph is a straight line.

60. Total distance, $s = 100 \text{ km}$

$$\text{Total time, } t = t_1 + t_2 = \frac{30 \text{ km}}{30 \text{ km/h}} + \frac{70 \text{ km}}{x \text{ (say)}}$$

$$t = 1 + \frac{70}{x} = \left(\frac{x+70}{x} \right) \text{ h}$$

$$\text{Now, average speed, } v = \frac{s}{t} = 40 \text{ km/h}$$

$$\text{According to question, } \frac{\frac{100}{\frac{x+70}{x}}}{\frac{x+70}{x}} = 40$$

$$\Rightarrow 10 = 4 \left(\frac{x+70}{x} \right)$$

$$\Rightarrow 10x = 4x + 280$$

$$\Rightarrow 6x = 280$$

$$\Rightarrow x = 46.67 \text{ km/h}$$

Speed for next 70 km to maintain average of 40 km/h is 46.67 km/h.

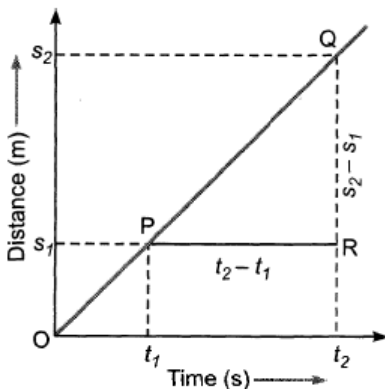
61. i. OA represented uniform acceleration and AB represented zero acceleration or constant velocity.

ii. The velocity of the body after the 20s is 20 m/s and after 80 s is zero or the body comes to rest.

iii. The negative acceleration of the body/retardation = $\frac{(0 - 20)}{(80 - 20)} = \frac{-20}{60} = -\frac{1}{3} \text{ m/s}^2$

iv. Distance covered by the body between 20th and 60th second = $(60 - 20) \times 20 \text{ m/s} = 40 \times 20 \text{ m/s} = 800 \text{ m}$.

62. Consider a body of mass 'm' moving with uniform velocity i.e. covering equal distance in equal interval of time. Graphically, it can be represented as follows:



From graph, Let a body covers distance s_1 and s_2 at times t_1 and t_2 respectively.

$$\text{Now, Slope of line PQ} = \tan \theta = \frac{QR}{PR} = \frac{s_2 - s_1}{t_2 - t_1} = \frac{\text{Displacement}}{\text{Time}}$$

As $\frac{\text{Displacement}}{\text{Time}}$ is velocity, so the slope of the distance-time graph gives velocity of the body.

63. Equation For Velocity-Time Relation:

Consider the velocity-time graph of an object that moves under uniform acceleration. The initial velocity of the object is u and then it increases to v in time t . The velocity changes at a uniform rate a . The perpendicular lines BC and BE are drawn from point B on the time and the velocity axes respectively, so that the initial velocity is represented by OA, the final velocity is represented by BC and the time interval t is represented by OC. $BD = BC - CD$, represents the change in velocity in time interval t .

Let us draw AD parallel to OC. From the graph, we observe that

$$BC = BD + DC = BD + OA$$

$$\text{Substituting } BC = v \text{ and } OA = u,$$

$$\text{we get } v = BD + u \dots (i)$$

$$\text{or } BD = v - u \dots (ii)$$

From the velocity-time graph,

The acceleration of the object is given by $a = \frac{\text{Change in velocity}}{\text{time taken}}$

$$= \frac{BD}{AD} = \frac{BD}{OC}$$

Substituting $OC = t$, we get

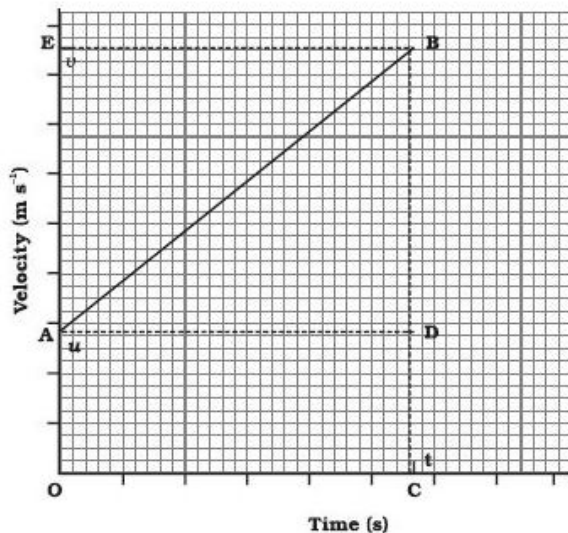
$$a = \frac{BD}{t}$$

$$BD = at \dots (iii)$$

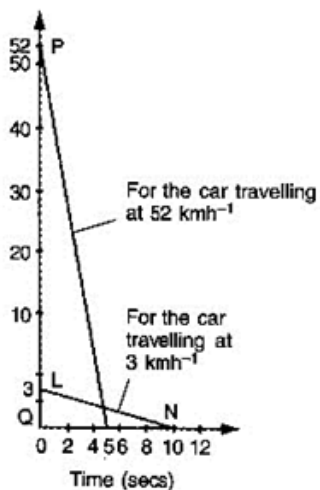
Using the equations (ii) & (iii). We get,

$$v = u + at$$

Chromium has a shiny appearance and does not corrode. It resists scratches. Chromium is expensive and it may not be economical to make the whole object out of chromium. So the object is made from a cheaper metal and only a coating of chromium over it is deposited. Jewellery makers electroplate silver and gold on less expensive metals. These ornaments have the appearance of silver or gold but are much less expensive.



64. i. Object dropped down from a height towards the surface of the earth.
 ii. Object thrown up with a velocity, retards uniformly.
 iii. Uniform circular motion, say planetary motion.
 iv. A bullet fired horizontally from a rifle, has acceleration in downward direction (due to gravity).
 v. A car moving with uniform velocity along a straight line.
65. The speed-time graph for both cars is shown below.



- i. Distance covered by car moving at 52 kmh^{-1}
 Or $52 \times \frac{5}{18} = 14.4 \text{ ms}^{-1}$
 $= \text{area of } \Delta PQR = \frac{1}{2} \times PQ \times QR = \frac{1}{2} \times 14.4 \times 5 = 36 \text{ m}$
- ii. Distance covered by car moving at 3 kmh^{-1}
 or $3 \times \frac{5}{18} = 0.83 \text{ ms}^{-1}$

$$= \text{area of } \triangle LNQ = \frac{1}{2} \times LQ \times QN = \frac{1}{2} \times 0.83 \times 10 = 4.15 \text{ m}$$

The car moving at 52 kmh^{-1} travels more distance on the application of brakes.

66. i. Reference point is important because it states the position of object correctly, as motion is relative in nature.
 ii. Yes, if other train is moving in adjacent line to the train we are sitting, then it seems that our train is moving in the opposite direction.
 iii. No, Odometer measures distance covered by a vehicle.
 iv. After half the circle, the particle will reach the diametrically opposite point, i.e. from point A to B.
 $\therefore$ Displacement after half circle = AB = r + r = 2r.
 v. If an object moves in a straight line from one point to another, then the magnitude of displacement and distance will be equal.
67. i. No, when the body comes back to the same position after travelling a distance, its displacement is zero though it has travelled some distance.
 ii. **Given:** Initial velocity $u = 0.7 \text{ m/s}$
 Final velocity, $v = 0$
 Acceleration, $a = -0.07 \text{ m/s}^2$
 Now, from the first equation of the motion,
 $v = u + at$
 $\Rightarrow 0 = 0.7 + (-0.07)t$
 $\Rightarrow 0.07t = 0.7$
 $\therefore t = \frac{0.7}{0.07} = 10 \text{ s}$
 Hence, the body will take 10 s to stop.

68. By using velocity- distance equation, $v^2 - u^2 = 2aS$.

For upward motion, acceleration due to gravity, $a = -g \text{ m/s}^2$.

Let the height attained by object = 'h'

Therefore, We have, $v^2 = u^2 - 2gh$ or $h = \frac{u^2 - v^2}{2g}$ (1)

But at highest point $v = 0$

Therefore, equation (1) become; $h = \frac{u^2}{2g}$

Now, For first ball, Height attained, $h_1 = \frac{u_1^2}{2g}$ (2)

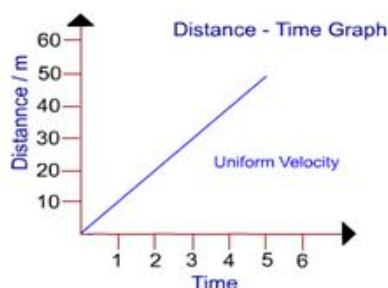
Similarly, For second ball, Height attained, $h_2 = \frac{u_2^2}{2g}$ (3)

Dividing equation, (2) by (3), we get.

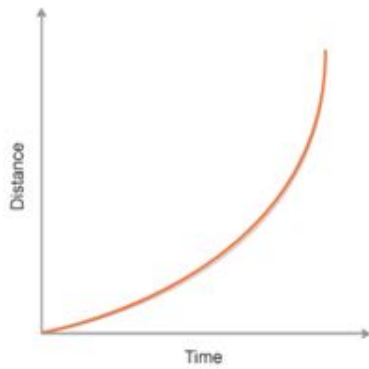
$$\frac{h_1}{h_2} = \frac{u_1^2/2g}{u_2^2/2g} = \frac{u_1^2}{u_2^2} \text{ or } h_1 : h_2 = u_1^2 : u_2^2 ; \text{ Hence proved.}$$

69. i. The motion of the body moving around a fixed point in a circular path is known as circular motion.
 ii. The diameter of the circular track is given as 200 m. That is $2r = 200 \text{ m}$.
 From this radius is calculated as Radius, $r = 100 \text{ m}$.
 In 40 sec the athlete completes one round. So, in 2 mins and 20 secs, that is, 140 sec the athlete will complete = $\frac{140}{40} = 3.5$ (three and a half) rounds.
 One round is considered as the circumference of the circular track.
 The distance covered in 140 sec = $2\pi r \times 3.5 = 2 \times 3.14 \times 100 \times 3.5 = 2200 \text{ m}$.
 For each complete round, the displacement is zero. Therefore for 3 complete rounds, the displacement will be zero.
 At the end of his motion, the athlete will be in the diametrically opposite position. That is, displacement = diameter = 200 m.
 Hence, the distance covered is 2200 m and the displacement is 200 m.

70. i. An object moving with uniform speed:



ii. The object moving with non-uniform speed:



71. a. When A starts his journey at 4 sec, B has already covered a distance of 857 m
 b. A travels faster than B because A starts his journey late but crosses B and covers more distance than B in the same time as B
 c. Speed of A = $\frac{\text{Distance covered}}{\text{time taken}}$
 Let at $t = 12$ min, distance covered = 3500 m
 $\frac{3500}{12} = 375$ m/min
 d. Speed of B = $\frac{\text{Distance covered}}{\text{time taken}}$
 $V_B = \frac{3000}{12} = 214$ m/min
 e. Speed of approach of A towards B = 375 m/min - 214 m/min = 161 m/min
 f. Speed of separation of A from B = 161 m/min.

72. After first 2 s; $u = 0$, $s = 20$ m and $t = 2$ s

Using distance formula;

$$s = ut + \frac{1}{2} at^2$$

$$\text{or } 20\text{m} = 0 + \frac{1}{2} a \times (2)^2 \text{ s}^2 \text{ or } a = 10 \text{ ms}^{-2}$$

$$\text{Final velocity after } 2\text{s} = u + at = 0 + 10 \text{ ms}^{-2} \times 2\text{s} = 20 \text{ ms}^{-1}$$

For the next 4 s; $u = 20$ m/s, $t = 4$ s and $s = 160$ m

Using distance formula;

$$s = ut + \frac{1}{2} at^2$$

$$\text{or } 160 \text{ m} = 20 \text{ ms}^{-1} \times 4 \text{ s} + \frac{1}{2} a \times 16 \text{ s}^2$$

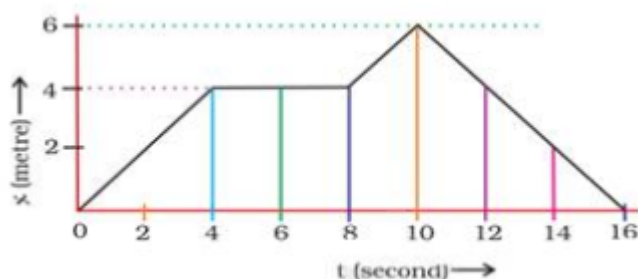
$$\text{or } 160 \text{ m} = 80 \text{ m} + a \times 8\text{s}^2$$

$$\text{or } a = \frac{80\text{m}}{8\text{s}^2} = 10 \text{ ms}^{-2}$$

Velocity after 7 s can be calculated as follows:

$$v = u + at = 0 + 10 \text{ ms}^{-2} \times 7\text{s} = 70 \text{ ms}^{-1}$$

73.



$$\text{Average velocity for first } 4\text{s} = \frac{\text{change in displacement}}{\text{time}}$$

$$= \frac{4\text{m} - 0\text{m}}{4\text{s}} = 1 \text{ ms}^{-1}$$

$$\text{Average velocity for next } 4\text{s} = \frac{4\text{m} - 4\text{m}}{8\text{s} - 4\text{s}} = \frac{0}{4} = 0 \text{ ms}^{-1}$$

$$\text{Average velocity for last } 6\text{s} = \frac{0 - 6\text{m}}{16\text{s} - 10\text{s}} = -1 \text{ ms}^{-1}$$

74. As given in the question that the insect takes 1 min to move from one point to a diametrically opposite point, which means in 1 min the insect will be completed a half revolution. So, the distance covered will be equal to 'half of the circumference' of the circular path and displacement will be equal to the diameter of the circular path.

Suppose the insect was at A initially and it moves along ACB to reach the diametrically opposite point B in 1 min.

i. $\therefore$ The distance moved in 1 min = πr

$$= 3.14 \times 10 = 31.4 \text{ cm}$$

ii. $\therefore$ Speed = $\frac{\text{Distance}}{\text{Time}} = \frac{31.4}{1} = 31.4 \text{ cm/min}$

iii. $\therefore$ Displacement, AB = $2r = 2 \times 10 = 20 \text{ cm}$

iv. $\therefore$ Average velocity,

$$v_{av} = \frac{\text{Displacement}}{\text{Time}} = \frac{20 \text{ cm}}{1 \text{ min}} = 20 \text{ cm/min.}$$

75. i. In the graph (iii), the distance between them will be same.

ii. In the graph (i), they are moving in opposite directions.

iii. Velocity of object A is positive in situation (i) because the slope of s-t graph is positive.

iv. Yes, in situations (i) and (ii) they are crossing each other because in these two cases, the s-t graph of A and B intersect each other.